The slope formula

1

a 3 **b** -3 **c** -1 **d** -1

e $\frac{3}{5}$ **f** -1 **g** 0 **h** 0

2 No. Reversing the order does not affect the final outcome of the slope. As long as you maintain the correct order by subtracting the corresponding y and x coordinates, the slope should be the same.

3

a Undefined **b** Undefined **c** Defined **d** Defined

e Undefined **f** Defined **g** Undefined **h** Defined

4

a $y = 25$ **b** 5

5

a $t = 3$ **b** $t = 15$ **c** $d = 20$ **d** $c = -\frac{12}{7}$

e $t = -23$

6 890 m

7

a 0 **b** 4 **c** No

8

a undefined **b** No **c** 1

9

a Neither **b** Undefined **c** 0 **d** Neither

e Neither **f** Neither **g** 0

10

a The change in x is equal to zero. The line is vertical.

b The line is horizontal. The change in y is equal to zero.

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11

a No

b Yes

c No

12 **D**

13

a 0

b Undefined

14 **B**

15 -20

Additional questions

16

a

i 14 **ii** 7 **iii** 2

b

i 12 **ii** 4 **iii** 3

c

i -12 **ii** 4 **iii** -3

d

i -6 **ii** 2 **iii** -3

17 Because a vertical line has no run. So the formula for the slope $\frac{\text{rise}}{\text{run}}$ will have a denominator of zero, and any fraction with a zero denominator is undefined.

18 0

19

a 2 **b** $-\frac{7}{5}$ **c** $\frac{1}{2}$ **d** $-\frac{2}{3}$

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20

a -2

b As x increases, the value of y decreases.

21

a 3 **b** $\frac{3}{5}$ **c** -4 **d** -3

e 0 **f** 0 **g** 1 **h** -1

22

a $\frac{5}{3}$ **b** $\frac{5}{3}$ **c** Yes

23 $m = 1$

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